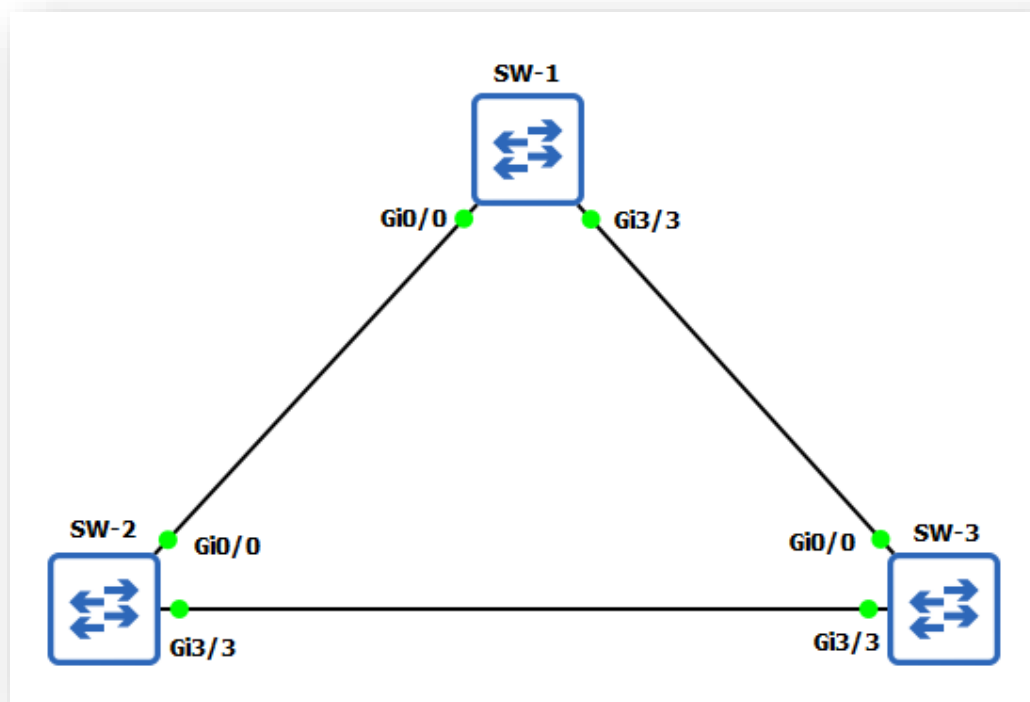


Spanning Tree Protocol (STP)

**Task**

1. Configure the SW-1 with IP 150.150.150.1, SW-2 with IP 150.150.150.2 and SW-3 with IP 150.150.150.3 for device management and hostnames.
2. Configure VTP on all switches. SW-1 as server and SW-2 & SW-3 as client
3. Create VLAN-10 as HR, VLAN-20 as Sale and VLAN-30 as Marketing.
4. Configure SW-1 as root for VLAN-1 & VLAN-10, SW-2 as root for VLAN-20 and SW-3 as root for VLAN-30.

Task-1: Configure Hostname on switches as SW-1, SW-2 and SW-3. Configure the SW-1 with IP 150.150.150.1, SW-2 with IP 150.150.150.2 and SW-3 with IP 150.150.150.3 for device management along with trunk links.

SW-1 Configuration:

```

Switch>en
Switch#config t
Switch(config)#hostname SW-1
SW-1(config)#no ip domain-lookup
SW-1(config)#int vlan1
SW-1(config-if)#ip address 150.150.150.1 255.255.255.0
SW-1(config-if)#no shut
SW-1(config-if)#exit
SW-1(config)#int gi0/0
SW-1(config-if)#switchport trunk encapsulation dot1q
SW-1(config-if)#switchport mode trunk
SW-1(config-if)#exit
  
```

```
SW-1(config)#int gi3/3
SW-1(config-if)#switchport trunk encapsulation dot1q
SW-1(config-if)#switchport mode trunk
SW-1(config-if)#exit
SW-1(config)#exit
SW-1#
```

SW-2 Configuration:

```
Switch>en
Switch#config t
Switch(config)#hostname SW-2
SW-2(config)#no ip domain-lookup
SW-2(config)#int vlan1
SW-2(config-if)#ip address 150.150.150.2 255.255.255.0
SW-2(config-if)#no shut
SW-2(config-if)#exit
SW-2(config)#int gi3/3
SW-2(config-if)#switchport trunk encapsulation dot1q
SW-2(config-if)#switchport mode trunk
SW-2(config-if)#exit
SW-2(config)#exit
SW-2#
```

SW-3 Configuration:

```
Switch>en
Switch#config t
Switch(config)#hostname SW-3
SW-3(config)#no ip domain-lookup
SW-3(config)#int vlan1
SW-3(config-if)#ip address 150.150.150.3 255.255.255.0
SW-3(config-if)#no shut
SW-3(config-if)#exit
SW-3(config)#exit
SW-3#
```

✓ Verification & Testing:

SW-1#sh int trunk

Port	Mode	Encapsulation	Status	Native vlan
Gi0/0	on	802.1q	trunking	1
Gi3/3	on	802.1q	trunking	1

Port	Vlans allowed on trunk
Gi0/0	1-4094
Gi3/3	1-4094

Port	Vlans allowed and active in management domain
Gi0/0	1

Gi3/3 1

Port Vlans in spanning tree forwarding state and not pruned

Gi0/0 1

Gi3/3 1

SW-2#sh int trunk

Port	Mode	Encapsulation	Status	Native vlan
Gi0/0	auto	n-802.1q	trunking	1
Gi3/3	on	802.1q	trunking	1

Port Vlans allowed on trunk

Gi0/0 1-4094

Gi3/3 1-4094

Port Vlans allowed and active in management domain

Gi0/0 1

Gi3/3 1

Port Vlans in spanning tree forwarding state and not pruned

Gi0/0 1

Gi3/3 none

SW-3#sh int trunk

Port	Mode	Encapsulation	Status	Native vlan
Gi0/0	auto	n-802.1q	trunking	1
Gi3/3	auto	n-802.1q	trunking	1

Port Vlans allowed on trunk

Gi0/0 1-4094

Gi3/3 1-4094

Port Vlans allowed and active in management domain

Gi0/0 1

Gi3/3 1

Port Vlans in spanning tree forwarding state and not pruned

Gi0/0 1

Gi3/3 1

Task-2: Configure VTP on all switches. SW-1 as server and SW-2 & SW-3 as client

SW-1 Configuration:

SW-1#config t

SW-1(config)#vtp mode server

Device mode already VTP Server for VLANS.

SW-1(config)#vtp domain rsatechforum.in

Changing VTP domain name from NULL to rsatechforum.in

```
SW-1(config)#vtp version 3
SW-1(config)#vtp password Pass@123
Setting device VTP password to Pass@123
SW-1(config)#exit
SW-1#
```

```
SW-1#vtp primary
```

This system is becoming primary server for feature vlan

```
No conflicting VTP3 devices found.
Do you want to continue? [confirm]
SW-1#
```

SW-2 Configuration:

```
SW-2#config t
SW-2(config)#vtp mode client
Setting device to VTP Client mode for VLANs.
SW-2(config)#vtp domain rsatechforum.in
Domain name already set to rsatechforum.in.
SW-2(config)#vtp version 3
SW-2(config)#vtp password Pass@123
Setting device VTP password to Pass@123
SW-2(config)#exit
SW-2#
```

SW-3 Configuration:

```
SW-3#config t
SW-3(config)#vtp mode client
Setting device to VTP Client mode for VLANs.
SW-3(config)#vtp domain rsatechforum.in
Changing VTP domain name from NULL to rsatechforum.in
SW-3(config)#vtp version 3
SW-3(config)#vtp password Pass@123
Setting device VTP password to Pass@123
SW-3(config)#exit
SW-3#
```

Task-3: Create VLAN-10 as HR, VLAN-20 as Sale and VLAN-30 as Marketing.

```
SW-1#config t
SW-1(config)#vlan 10
SW-1(config-vlan)#name HR
SW-1(config-vlan)#exit
SW-1(config)#vlan 20
SW-1(config-vlan)#name Sale
SW-1(config-vlan)#exit
SW-1(config)#vlan 30
SW-1(config-vlan)#name Marketing
SW-1(config-vlan)#exit
SW-1(config)#exit
```

SW-1#

✓ **Verification & Testing:****SW-1#sh vlan brief**

VLAN Name	Status	Ports
1 default	active	Gi0/1, Gi0/2, Gi0/3, Gi1/0 Gi1/1, Gi1/2, Gi1/3, Gi2/0 Gi2/1, Gi2/2, Gi2/3, Gi3/0 Gi3/1, Gi3/2
10 HR	active	
20 Sale	active	
30 Marketing	active	
1002 fddi-default	act/unsup	
1003 trcrf-default	act/unsup	
1004 fddinet-default	act/unsup	
1005 trbrf-default	act/unsup	

SW-1#

SW-1#sh vtp status

VTP Version capable : 1 to 3
VTP version running : 3
 VTP Domain Name : rsatechforum.in
 VTP Pruning Mode : Disabled
 VTP Traps Generation : Disabled
 Device ID : 0c17.9259.8000

Feature VLAN:

VTP Operating Mode : Primary Server

Number of existing VLANs : 8
 Number of existing extended VLANs: 0
 Maximum VLANs supported locally : 4096
 Configuration Revision : 3
 Primary ID : 0c17.9259.8000
 Primary Description : SW-1
 MD5 digest : 0x2B 0x55 0xC4 0x96 0xFD 0x1B 0x5A 0x66
 0x24 0xB2 0x4F 0x08 0x40 0xE5 0x8F 0x7D

Feature MST:

VTP Operating Mode : Transparent

Feature UNKNOWN:

VTP Operating Mode : Transparent

SW-2#sh vlan brief

VLAN Name	Status	Ports
1 default	active	Gi0/1, Gi0/2, Gi0/3, Gi1/0 Gi1/1, Gi1/2, Gi1/3, Gi2/0 Gi2/1, Gi2/2, Gi2/3, Gi3/0 Gi3/1, Gi3/2
10 HR	active	
20 Sale	active	
30 Marketing	active	
1002 fddi-default	act/unsup	
1003 trcrf-default	act/unsup	
1004 fddinet-default	act/unsup	
1005 trbrf-default	act/unsup	

SW-2#

SW-2#sh vtp status

VTP Version capable : 1 to 3
VTP version running : 3
 VTP Domain Name : rsatechforum.in
 VTP Pruning Mode : Disabled
 VTP Traps Generation : Disabled
 Device ID : 0c17.92f5.8000

Feature VLAN:

VTP Operating Mode : Client
 Number of existing VLANs : 8
 Number of existing extended VLANs: 0
 Maximum VLANs supported locally : 4096
 Configuration Revision : 3
 Primary ID : 0c17.9259.8000
 Primary Description : SW-1
 MD5 digest : 0x2B 0x55 0xC4 0x96 0xFD 0x1B 0x5A 0x66
 0x24 0xB2 0x4F 0x08 0x40 0xE5 0x8F 0x7D

Feature MST:

VTP Operating Mode : Transparent

Feature UNKNOWN:

VTP Operating Mode : Transparent

SW-2#

SW-3#sh vlan brief

VLAN Name	Status	Ports
1 default	active	Gi0/1, Gi0/2, Gi0/3, Gi1/0

Gi1/1, Gi1/2, Gi1/3, Gi2/0
 Gi2/1, Gi2/2, Gi2/3, Gi3/0
 Gi3/1, Gi3/2

```
10 HR active
20 Sale active
30 Marketing active
1002 fddi-default act/unsup
1003 trcrf-default act/unsup
1004 fddinet-default act/unsup
1005 trbrf-default act/unsup
SW-3#
```

SW-3#sh vtp status

```
VTP Version capable : 1 to 3
VTP version running : 3
VTP Domain Name : rsatechforum.in
VTP Pruning Mode : Disabled
VTP Traps Generation : Disabled
Device ID : 0c17.929f.8000
```

Feature VLAN:

```
-----
VTP Operating Mode : Client
Number of existing VLANs : 8
Number of existing extended VLANs : 0
Maximum VLANs supported locally : 4096
Configuration Revision : 3
Primary ID : 0c17.9259.8000
Primary Description : SW-1
MD5 digest : 0x2B 0x55 0xC4 0x96 0xFD 0x1B 0x5A 0x66
              0x24 0xB2 0x4F 0x08 0x40 0xE5 0x8F 0x7D
```

Feature MST:

```
-----
VTP Operating Mode : Transparent
```

Feature UNKNOWN:

```
-----
VTP Operating Mode : Transparent
```

SW-3#

SW-1#sh spanning-tree**VLAN0001**

Spanning tree enabled protocol ieee

Root ID Priority 32769

Address 0c17.9259.7100

This bridge is the root

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)

Address 0c17.9259.7100
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.Nbr	Type
Gi0/0	Desg	FWD	4	128.1	P2p
Gi0/1	Desg	FWD	4	128.2	P2p
Gi0/2	Desg	FWD	4	128.3	P2p
Gi0/3	Desg	FWD	4	128.4	P2p
Gi1/0	Desg	FWD	4	128.5	P2p
Gi1/1	Desg	FWD	4	128.6	P2p
Gi1/2	Desg	FWD	4	128.7	P2p
Gi1/3	Desg	FWD	4	128.8	P2p

Interface	Role	Sts	Cost	Prio.Nbr	Type
Gi2/0	Desg	FWD	4	128.9	P2p
Gi2/1	Desg	FWD	4	128.10	P2p
Gi2/2	Desg	FWD	4	128.11	P2p
Gi2/3	Desg	FWD	4	128.12	P2p
Gi3/0	Desg	FWD	4	128.13	P2p
Gi3/1	Desg	FWD	4	128.14	P2p
Gi3/2	Desg	FWD	4	128.15	P2p
Gi3/3	Desg	FWD	4	128.16	P2p

VLAN0010

Spanning tree enabled protocol ieee

Root ID Priority 32778

Address 0c17.9259.7100

This bridge is the root

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32778 (priority 32768 sys-id-ext 10)

Address 0c17.9259.7100

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.Nbr	Type
Gi0/0	Desg	FWD	4	128.1	P2p
Gi3/3	Desg	FWD	4	128.16	P2p

VLAN0020

Spanning tree enabled protocol ieee

Root ID Priority 32788

Address 0c17.9259.7100

This bridge is the root

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32788 (priority 32768 sys-id-ext 20)
 Address 0c17.9259.7100
 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
 Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.	Nbr	Type
Gi0/0	Desg	FWD	4	128.1		P2p
Gi3/3	Desg	FWD	4	128.16		P2p

VLAN0030

Spanning tree enabled protocol ieee

Root ID Priority 32798

Address 0c17.9259.7100

This bridge is the root

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32798 (priority 32768 sys-id-ext 30)
 Address 0c17.9259.7100
 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
 Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.	Nbr	Type
Gi0/0	Desg	FWD	4	128.1		P2p
Gi3/3	Desg	FWD	4	128.16		P2p

SW-1#sh spanning-tree summary

Switch is in pvst mode
 Root bridge for: VLAN0001, VLAN0010, VLAN0020, VLAN0030
 Extended system ID is enabled
 Portfast Default is disabled
 Portfast Edge BPDU Guard Default is disabled
 Portfast Edge BPDU Filter Default is disabled
 Loopguard Default is disabled
 PVST Simulation Default is enabled but inactive in pvst mode
 Bridge Assurance is enabled but inactive in pvst mode
 EtherChannel misconfig guard is enabled
 Configured Pathcost method used is short
 UplinkFast is disabled
 BackboneFast is disabled

Name	Blocking	Listening	Learning	Forwarding	STP Active
VLAN0001	0	0	0	16	16
VLAN0010	0	0	0	2	2
VLAN0020	0	0	0	2	2
VLAN0030	0	0	0	2	2
4 vlans	0	0	0	22	22

SW-2#sh spanning-tree

VLAN0001

Spanning tree enabled protocol ieee

Root ID Priority 32769

Address 0c17.9259.7100

Cost 4

Port 1 (GigabitEthernet0/0)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)

Address 0c17.92f5.9600

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.Nbr	Type
Gi0/0	Root	FWD	4	128.1	P2p
Gi0/1	Desg	FWD	4	128.2	P2p
Gi0/2	Desg	FWD	4	128.3	P2p
Gi0/3	Desg	FWD	4	128.4	P2p
Gi1/0	Desg	FWD	4	128.5	P2p
Gi1/1	Desg	FWD	4	128.6	P2p
Gi1/2	Desg	FWD	4	128.7	P2p

Interface	Role	Sts	Cost	Prio.Nbr	Type
Gi1/3	Desg	FWD	4	128.8	P2p
Gi2/0	Desg	FWD	4	128.9	P2p
Gi2/1	Desg	FWD	4	128.10	P2p
Gi2/2	Desg	FWD	4	128.11	P2p
Gi2/3	Desg	FWD	4	128.12	P2p
Gi3/0	Desg	FWD	4	128.13	P2p
Gi3/1	Desg	FWD	4	128.14	P2p
Gi3/2	Desg	FWD	4	128.15	P2p
Gi3/3	Altn	BLK	4	128.16	P2p

VLAN0010

Spanning tree enabled protocol ieee

Root ID Priority 32778

Address 0c17.9259.7100

Cost 4

Port 1 (GigabitEthernet0/0)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32778 (priority 32768 sys-id-ext 10)

Address 0c17.92f5.9600

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.Nbr	Type
-----------	------	-----	------	----------	------

```

Gi0/0      Root FWD 4      128.1  P2p
Gi3/3      Altn BLK 4      128.16 P2p

```

VLAN0020

Spanning tree enabled protocol ieee

Root ID Priority 32788

Address 0c17.9259.7100

Cost 4

Port 1 (GigabitEthernet0/0)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32788 (priority 32768 sys-id-ext 20)

Address 0c17.92f5.9600

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.Nbr	Type
Gi0/0	Root FWD	4	128.1	P2p	
Gi3/3	Altn BLK	4	128.16	P2p	

```

Gi0/0      Root FWD 4      128.1  P2p
Gi3/3      Altn BLK 4      128.16 P2p

```

VLAN0030

Spanning tree enabled protocol ieee

Root ID Priority 32798

Address 0c17.9259.7100

Cost 4

Port 1 (GigabitEthernet0/0)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32798 (priority 32768 sys-id-ext 30)

Address 0c17.92f5.9600

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.Nbr	Type
Gi0/0	Root FWD	4	128.1	P2p	
Gi3/3	Altn BLK	4	128.16	P2p	

```

Gi0/0      Root FWD 4      128.1  P2p
Gi3/3      Altn BLK 4      128.16 P2p

```

SW-2#sh spanning-tree summary

Switch is in pvst mode

Root bridge for : none

Extended system ID is enabled

Portfast Default is disabled

Portfast Edge BPDU Guard Default is disabled

Portfast Edge BPDU Filter Default is disabled

Loopguard Default is disabled

PVST Simulation Default is enabled but inactive in pvst mode

Bridge Assurance is enabled but inactive in pvst mode

EtherChannel misconfig guard is enabled
 Configured Pathcost method used is short
 UplinkFast is disabled
 BackboneFast is disabled

Name	Blocking	Listening	Learning	Forwarding	STP Active
------	----------	-----------	----------	------------	------------

VLAN0001	1	0	0	15	16
VLAN0010	1	0	0	1	2
VLAN0020	1	0	0	1	2
VLAN0030	1	0	0	1	2

4 vlans	4	0	0	18	22
---------	---	---	---	----	----

SW-3#sh spanning-tree

VLAN0001

Spanning tree enabled protocol ieee

Root ID Priority 32769

Address 0c17.9259.7100

Cost 4

Port 1 (GigabitEthernet0/0)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)

Address 0c17.929f.7d00

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.Nbr	Type
-----------	------	-----	------	----------	------

Gi0/0	Root	FWD	4	128.1	P2p
Gi0/1	Desg	FWD	4	128.2	P2p
Gi0/2	Desg	FWD	4	128.3	P2p
Gi0/3	Desg	FWD	4	128.4	P2p
Gi1/0	Desg	FWD	4	128.5	P2p
Gi1/1	Desg	FWD	4	128.6	P2p
Gi1/2	Desg	FWD	4	128.7	P2p

Interface	Role	Sts	Cost	Prio.Nbr	Type
-----------	------	-----	------	----------	------

Gi1/3	Desg	FWD	4	128.8	P2p
Gi2/0	Desg	FWD	4	128.9	P2p
Gi2/1	Desg	FWD	4	128.10	P2p
Gi2/2	Desg	FWD	4	128.11	P2p
Gi2/3	Desg	FWD	4	128.12	P2p
Gi3/0	Desg	FWD	4	128.13	P2p
Gi3/1	Desg	FWD	4	128.14	P2p
Gi3/2	Desg	FWD	4	128.15	P2p
Gi3/3	Desg	FWD	4	128.16	P2p

VLAN0010

Spanning tree enabled protocol ieee

Root ID Priority 32778

Address 0c17.9259.7100

Cost 4

Port 1 (GigabitEthernet0/0)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32778 (priority 32768 sys-id-ext 10)

Address 0c17.929f.7d00

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.	Nbr	Type
Gi0/0	Root	FWD	4	128.1		P2p
Gi3/3	Desg	FWD	4	128.16		P2p

VLAN0020

Spanning tree enabled protocol ieee

Root ID Priority 32788

Address 0c17.9259.7100

Cost 4

Port 1 (GigabitEthernet0/0)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32788 (priority 32768 sys-id-ext 20)

Address 0c17.929f.7d00

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.	Nbr	Type
Gi0/0	Root	FWD	4	128.1		P2p
Gi3/3	Desg	FWD	4	128.16		P2p

VLAN0030

Spanning tree enabled protocol ieee

Root ID Priority 32798

Address 0c17.9259.7100

Cost 4

Port 1 (GigabitEthernet0/0)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32798 (priority 32768 sys-id-ext 30)

Address 0c17.929f.7d00

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.	Nbr	Type
-----------	------	-----	------	-------	-----	------

```

Gi0/0      Root FWD 4      128.1   P2p
Gi3/3      Desg FWD 4      128.16  P2p
  
```

SW-3#sh spanning-tree summary

```

Switch is in pvst mode
Root bridge for : none
Extended system ID is enabled
Portfast Default is disabled
Portfast Edge BPDU Guard Default is disabled
Portfast Edge BPDU Filter Default is disabled
Loopguard Default is disabled
PVST Simulation Default is enabled but inactive in pvst mode
Bridge Assurance is enabled but inactive in pvst mode
EtherChannel misconfig guard is enabled
Configured Pathcost method used is short
UplinkFast is disabled
BackboneFast is disabled
  
```

Name	Blocking	Listening	Learning	Forwarding	STP Active
VLAN0001	0	0	0	16	16
VLAN0010	0	0	0	2	2
VLAN0020	0	0	0	2	2
VLAN0030	0	0	0	2	2
4 vlans	0	0	0	22	22

Task-4: Configure SW-1 as root for VLAN-1 & VLAN-10, SW-2 as root for VLAN-20 and SW-3 as root for VLAN-30.

SW-1 Configuration:

```

SW-1(config)#spanning-tree vlan 1,10 root primary
SW-1(config)#exit
SW-1#
  
```

SW-2 Configuration:

```

SW-2(config)#spanning-tree vlan 20 root primary
SW-2(config)#exit
SW-2#
  
```

SW-3 Configuration:

```

SW-3(config)#spanning-tree vlan 30 priority 4096
SW-3(config)#exit
SW-3#
  
```

✓ **Verification & Testing:****SW-1#sh spanning-tree****VLAN0001**

Spanning tree enabled protocol ieee

Root ID Priority 24577

Address 0c17.9259.7100

This bridge is the root

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 24577 (priority 24576 sys-id-ext 1)

Address 0c17.9259.7100

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.Nbr	Type
-----------	------	-----	------	----------	------

Gi0/0	Desg	FWD	4	128.1	P2p
Gi0/1	Desg	FWD	4	128.2	P2p
Gi0/2	Desg	FWD	4	128.3	P2p
Gi0/3	Desg	FWD	4	128.4	P2p
Gi1/0	Desg	FWD	4	128.5	P2p
Gi1/1	Desg	FWD	4	128.6	P2p
Gi1/2	Desg	FWD	4	128.7	P2p
Gi1/3	Desg	FWD	4	128.8	P2p

Interface	Role	Sts	Cost	Prio.Nbr	Type
-----------	------	-----	------	----------	------

Gi2/0	Desg	FWD	4	128.9	P2p
Gi2/1	Desg	FWD	4	128.10	P2p
Gi2/2	Desg	FWD	4	128.11	P2p
Gi2/3	Desg	FWD	4	128.12	P2p
Gi3/0	Desg	FWD	4	128.13	P2p
Gi3/1	Desg	FWD	4	128.14	P2p
Gi3/2	Desg	FWD	4	128.15	P2p
Gi3/3	Desg	FWD	4	128.16	P2p

VLAN0010

Spanning tree enabled protocol ieee

Root ID Priority 24586

Address 0c17.9259.7100

This bridge is the root

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 24586 (priority 24576 sys-id-ext 10)

Address 0c17.9259.7100

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.Nbr	Type
-----------	------	-----	------	----------	------

Gi0/0	Desg	FWD	4	128.1	P2p
Gi3/3	Desg	FWD	4	128.16	P2p

VLAN0020

Spanning tree enabled protocol ieee

Root ID Priority 24596

Address 0c17.92f5.9600

Cost 4

Port 1 (GigabitEthernet0/0)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32788 (priority 32768 sys-id-ext 20)

Address 0c17.9259.7100

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

SW-2#sh spanning-tree

VLAN0001

Spanning tree enabled protocol ieee

Root ID Priority 24577

Address 0c17.9259.7100

Cost 4

Port 1 (GigabitEthernet0/0)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)

Address 0c17.92f5.9600

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.Nbr	Type
-----------	------	-----	------	----------	------

Gi0/0	Root	FWD	4	128.1	P2p
Gi0/1	Desg	FWD	4	128.2	P2p
Gi0/2	Desg	FWD	4	128.3	P2p
Gi0/3	Desg	FWD	4	128.4	P2p
Gi1/0	Desg	FWD	4	128.5	P2p
Gi1/1	Desg	FWD	4	128.6	P2p
Gi1/2	Desg	FWD	4	128.7	P2p

Interface	Role	Sts	Cost	Prio.Nbr	Type
-----------	------	-----	------	----------	------

Gi1/3	Desg	FWD	4	128.8	P2p
Gi2/0	Desg	FWD	4	128.9	P2p
Gi2/1	Desg	FWD	4	128.10	P2p
Gi2/2	Desg	FWD	4	128.11	P2p
Gi2/3	Desg	FWD	4	128.12	P2p


```

Gi3/0      Desg FWD 4      128.13 P2p
Gi3/1      Desg FWD 4      128.14 P2p
Gi3/2      Desg FWD 4      128.15 P2p
Gi3/3      Altn BLK 4      128.16 P2p

```

VLAN0010

Spanning tree enabled protocol ieee

Root ID Priority 24586

Address 0c17.9259.7100

Cost 4

Port 1 (GigabitEthernet0/0)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32778 (priority 32768 sys-id-ext 10)

Address 0c17.92f5.9600

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.Nbr	Type
Gi0/0	Root	FWD	4	128.1	P2p
Gi3/3	Altn	BLK	4	128.16	P2p

VLAN0020

Spanning tree enabled protocol ieee

Root ID Priority 24596

Address 0c17.92f5.9600

This bridge is the root

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 24596 (priority 24576 sys-id-ext 20)

Address 0c17.92f5.9600

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.Nbr	Type
Gi0/0	Desg	FWD	4	128.1	P2p
Gi3/3	Desg	FWD	4	128.16	P2p

VLAN0030

Spanning tree enabled protocol ieee

Root ID Priority 4126

Address 0c17.929f.7d00

Cost 4

Port 16 (GigabitEthernet3/3)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32798 (priority 32768 sys-id-ext 30)

Address 0c17.92f5.9600

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.	Nbr	Type
Gi0/0	Altn	BLK	4	128.1		P2p
Gi3/3	Root	FWD	4	128.16		P2p

SW-3#sh spanning-tree

VLAN0001

Spanning tree enabled protocol ieee

Root ID Priority 24577

Address 0c17.9259.7100

Cost 4

Port 1 (GigabitEthernet0/0)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)

Address 0c17.929f.7d00

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.	Nbr	Type
Gi0/0	Root	FWD	4	128.1		P2p
Gi0/1	Desg	FWD	4	128.2		P2p
Gi0/2	Desg	FWD	4	128.3		P2p
Gi0/3	Desg	FWD	4	128.4		P2p
Gi1/0	Desg	FWD	4	128.5		P2p
Gi1/1	Desg	FWD	4	128.6		P2p
Gi1/2	Desg	FWD	4	128.7		P2p

Interface	Role	Sts	Cost	Prio.	Nbr	Type
Gi1/3	Desg	FWD	4	128.8		P2p
Gi2/0	Desg	FWD	4	128.9		P2p
Gi2/1	Desg	FWD	4	128.10		P2p
Gi2/2	Desg	FWD	4	128.11		P2p
Gi2/3	Desg	FWD	4	128.12		P2p
Gi3/0	Desg	FWD	4	128.13		P2p
Gi3/1	Desg	FWD	4	128.14		P2p
Gi3/2	Desg	FWD	4	128.15		P2p
Gi3/3	Desg	FWD	4	128.16		P2p

VLAN0010

Spanning tree enabled protocol ieee

Root ID Priority 24586

Address 0c17.9259.7100

Cost 4

Port 1 (GigabitEthernet0/0)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32778 (priority 32768 sys-id-ext 10)
 Address 0c17.929f.7d00
 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
 Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.	Nbr	Type
Gi0/0	Root	FWD	4	128.1		P2p
Gi3/3	Desg	FWD	4	128.16		P2p

VLAN0020

Spanning tree enabled protocol ieee

Root ID Priority 24596

Address 0c17.92f5.9600

Cost 4

Port 16 (GigabitEthernet3/3)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32788 (priority 32768 sys-id-ext 20)
 Address 0c17.929f.7d00
 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
 Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.	Nbr	Type
Gi0/0	Altn	BLK	4	128.1		P2p
Gi3/3	Root	FWD	4	128.16		P2p

VLAN0030

Spanning tree enabled protocol ieee

Root ID Priority 4126

Address 0c17.929f.7d00

This bridge is the root

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 4126 (priority 4096 sys-id-ext 30)
 Address 0c17.929f.7d00
 Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
 Aging Time 300 sec

Interface	Role	Sts	Cost	Prio.	Nbr	Type
Gi0/0	Desg	FWD	4	128.1		P2p
Gi3/3	Desg	FWD	4	128.16		P2p

SW-1#sh spanning-tree interface gigabitEthernet 0/0

Vlan	Role	Sts	Cost	Prio.	Nbr	Type
------	------	-----	------	-------	-----	------

VLAN0001	Desg	FWD	4	128.1		P2p
VLAN0010	Desg	FWD	4	128.1		P2p
VLAN0020	Root	FWD	4	128.1		P2p
VLAN0030	Desg	FWD	4	128.1		P2p

SW-1#

SW-1#sh spanning-tree interface gigabitEthernet 3/3

Vlan	Role	Sts	Cost	Prio.	Nbr	Type
------	------	-----	------	-------	-----	------

VLAN0001	Desg	FWD	4	128.16		P2p
VLAN0010	Desg	FWD	4	128.16		P2p
VLAN0020	Desg	FWD	4	128.16		P2p
VLAN0030	Root	FWD	4	128.16		P2p

SW-1#

SW-2#sh spanning-tree interface gigabitEthernet 0/0

Vlan	Role	Sts	Cost	Prio.	Nbr	Type
------	------	-----	------	-------	-----	------

VLAN0001	Root	FWD	4	128.1		P2p
VLAN0010	Root	FWD	4	128.1		P2p
VLAN0020	Desg	FWD	4	128.1		P2p
VLAN0030	Altn	BLK	4	128.1		P2p

SW-2#

SW-2#sh spanning-tree interface gigabitEthernet 3/3

Vlan	Role	Sts	Cost	Prio.	Nbr	Type
------	------	-----	------	-------	-----	------

VLAN0001	Altn	BLK	4	128.16		P2p
VLAN0010	Altn	BLK	4	128.16		P2p
VLAN0020	Desg	FWD	4	128.16		P2p
VLAN0030	Root	FWD	4	128.16		P2p

SW-2#

SW-3#sh spanning-tree interface gigabitEthernet 0/0

Vlan	Role	Sts	Cost	Prio.	Nbr	Type
------	------	-----	------	-------	-----	------

VLAN0001	Root	FWD	4	128.1		P2p
VLAN0010	Root	FWD	4	128.1		P2p
VLAN0020	Altn	BLK	4	128.1		P2p
VLAN0030	Desg	FWD	4	128.1		P2p

SW-3#

SW-3#sh spanning-tree interface gigabitEthernet 3/3

Vlan	Role	Sts	Cost	Prio.	Nbr	Type
VLAN0001	Desg	FWD	4	128.16		P2p
VLAN0010	Desg	FWD	4	128.16		P2p
VLAN0020	Root	FWD	4	128.16		P2p
VLAN0030	Desg	FWD	4	128.16		P2p

SW-3#

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Important Commands:

```
sh int trunk  
sh vlan brief  
sh vtp status  
sh spanning-tree  
sh spanning-tree summary  
sh spanning-tree interface gigabitEthernet x/x
```

Ratan